Name of the Student..... **Scholar Number.....**

Maulana Azad National Institute of Technology, BHOPAL

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

EXAMINATION: MID TERM **MONTH and YEAR: Nov 2025**

Course: M. Tech **Semester:** I **Branch: CN/AI**

Subject Code: CN-1103/AI-1163 **Subject Name: Wireless Networks**

Maximum Marks: 20 marks

Duration: 60 minutes **Date:** 06-11-2025 **Time:** 09:30 – 10:30 am

Note: 1. Attempt all questions, **2.** Assume the necessary data if required and mention it clearly.

Q. No.	Questions	Marks
1	Discuss the evolution of wireless communication technologies from 1G to 6G, focusing on the key enabling technologies, data rate enhancements, and modulation/multiple access techniques in each generation. How have these developments transformed wireless connectivity from voice-only communication to ultra-reliable, low-latency, AI-driven systems?	**(8)**
2	A transmitter radiates 20 W of power at a carrier frequency of 2.4 GHz through a unity-gain antenna. (a) Express the transmit power in dBm and dBW. (b) Compute the received power (in dBm) at a distance of 200 m and 10 km under free-space conditions. How much is the difference in dBm when the distance changes?	**(8)**
3	In a cellular system, the network operator has chosen a cluster size of N = 7 to balance capacity and coverage. If the system operates in an environment with a path loss exponent of **n = 4**, and we consider the six nearest co-channel interferers: (a) Calculate the co-channel reuse ratio. (b) Determine the Signal-to-Interference Ratio (SIR) in dB at the cell edge (worst-case scenario). (c) Based on your result in (b), would this cluster size be acceptable if the minimum required (SIR) is 18 dB?	**(8)**
4.	Explain the concept of traffic engineering in cellular systems. Cellular network involves critical trade-offs between cost and service quality. Explain the economic and quality trade-off in cellular traffic engineering, using Grade of Service **(GoS)** and total system capacity as the key metrics.	**(8)**
5.	Write short notes of two: - (i) Mobile Radio propagation (ii) Media-access techniques (iii) Channel Assignment.	**(8)**

**********END**********

Name of the Student..... **Scholar Number.** 2S21201110

Maulana Azad National Institute of Technology, BHOPAL

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

EXAMINATION: END TERM **MONTH and YEAR: Dec 2025**

Course: M. Tech **Semester:** I **Branch: CN/AI**

Subject Code: CN-1103/AI-1163 **Subject Name: Wireless Networks**

Maximum Marks: 50 marks

Duration: 2 Hours **Date:** 11-12-2025 **Time:** 09:30 – 11:30 am

Note: 1. Attempt all questions, **2.** Assume the necessary data if required and mention it clearly.

Q. No.	Questions	Marks
1 (a)	Explain how inter-cell interference, intra-cell interference, and adjacent-channel interference impact Wireless network performance.	(5)
(b)	A cellular operator has a total system bandwidth of 60 MHz. Each simplex RF channel occupies 100 kHz of bandwidth. The network uses a cluster size of $N = 4$ *and* 7 for frequency reuse. (a) Determine the number of full-duplex traffic channels available per cell. (b) Using the Erlang B model with a target blocking probability (GoS) of 1%, calculate the traffic capacity (in Erlangs) per cell.	(5)
2 (a)	What are the characteristics of normal and abnormal handoffs? A mobile user travels at 50 km/h through a cell with a radius of 1 km. The dwell time is defined as the time the user remains within the cell boundary. • Calculate the dwell time for the mobile user in the cell. • If a handoff requires 5 seconds to complete, determine if there is sufficient time for a seamless handoff.	(5)
(b)	A 5G base station transmits a power of 30 dBm at a carrier frequency of 28 GHz using an antenna with a gain of 20 dBi. The mobile user's antenna has a gain of 5 dBi. (a) Using the free-space path loss (FSPL) model, calculate the received power at a distance of 100 meters. (b) Repeat the calculation for a lower carrier frequency of 3.5 GHz, keeping the transmit power, antenna gains, and distance the same. (c) Explain why mmWave frequencies (such as 28 GHz) require techniques like beamforming and dense base-station deployment to achieve reliable communication.	(5)
3 (a)	Explain MAC Protocols in IEEE 802.11 Wireless LAN. How are DCF and PCF used for collision avoidance?	(5)
(b)	Design a Cognitive Radio Network for a smart city environment supporting IoT devices. Describe components of: • spectrum sensing	(5)

	• spectrum management/decision • spectrum mobility • Spectrum sharing/interference avoidance.	
4. (a) (b)	Describe the architecture of 5G O-RAN. How enhanced mobile broadband, massive IoT, and ultra-reliable low-latency communication are achieved in 5G networks. What are the different medium access techniques? Consider a CDMA system in which user A & B have the code c_A= <1,-1,-1, 1, -1, 1> and c_B= < 1, 1, -1, -1, 1, 1> respectively. (i) Show the output of the receiver if A transmits data bit 1 in fist case and & 0 next. (ii) Show the output of the receiver if A tries to read the B's data.	(5) (5)
5.	Write short notes of two: - (i) OFDM (ii) 6G Future Networks (iii) Channel Assignment.	(10)

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